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Education

University of Wisconsin–Madison

Expected May 2027

B.S. Computer Science

GPA: 3.7/4.0

Coursework: Algorithms, Computer Vision, Data Management for Data Science, Mobile Systems, Optimization

Technical Skills

Languages: Python, C, C++, Rust, TypeScript/JavaScript, Swift, Kotlin, Go, SQL

ML & Inference: MLX, PyTorch, llama.cpp, safetensors, Apple Accelerate

Systems & GPU: tokio, proptest, LLDB, AddressSanitizer, Metal, WebGPU/WGSL, SwiftUI

Projects

Qwen3.8-Flash-Next NVMe Inference Engine [GitHub]

Aug 2026 – Present

- Built an inference engine that runs the **180B-parameter** Qwen3.8-Flash-Next MoE on a 36 GB Mac by streaming 4-bit experts from NVMe through an adaptive cache, sustaining **1.33 tok/s** decode
- Porting the runtime to C with Accelerate GEMM kernels and a byte-level BPE tokenizer that matches Hugging Face token for token on **416/416** test strings

spack: Local Coding Agent [GitHub]

Aug 2026 – Present

- Built a Rust coding agent that fixes bugs with a local Qwen3.8-27B through llama.cpp, using sandboxed read, edit, search, and shell tools behind approval gates; **0 malformed** tool calls across 1,297
- Wrote a 21-repo bug-fix benchmark whose prompts never name the bug; the local 27B solves **90%** of runs, vs 19% for gpt-4o-mini and 95% for gpt-5-mini
- Added a grader that checks every test file against the original, which caught gpt-4o-mini rewriting or deleting tests in **20 of 126** runs; rerun under it, its score fell from 40% to 19%

Learned Expert Prefetching for MoE Inference [GitHub]

Oct 2026 – Present

- Traced routing and hidden states for all 48 MoE layers over 30 prompts, then trained low-rank per-layer heads that pick the next layer's experts with **67%** top-10 recall, vs 22% for the engine's heuristic
- Hooked the predictors into the engine's read path with pin-until-used caching, raising prefetch hit rate from under 1% to **55%** and cutting median expert wait 65% with bit-identical outputs on 8 held-out prompts
- Built a trace-driven cache simulator and ran ablations on the real model, finding that without pinning the cache evicts prefetched experts before use and hit rate collapses to 8%

Interactive 3D Fluid Simulator (MLS-MPM) [Demo] [GitHub]

Sep 2025–Aug 2026

- Wrote a real-time 3D fluid simulator in WebGPU that runs **400,000 particles at 75+ fps** in the browser on an M3 Max, with all particle and grid state resident on the GPU across six compute kernels per substep
- Worked around WGSL's lack of float atomics by scattering particle momentum to the grid as fixed-point integer atomics; added a dam break, moving piston, particle injection, and an orbit camera

Zwift Shifter (macOS Virtual Gear Shifter) [GitHub] [Blog]

Aug 2026

- Reverse-engineered the Zwift Click v2 Bluetooth protocol from Zwift's logs and LLDB, isolating the **2 bits** in its 32-bit button map that shift gears up and down
- Shipped a SwiftUI app that injects a dylib into Zwift, which feeds 7-byte button packets to its Bluetooth bridge with **80 ms** press and release timing so each click is exactly one shift

Additional

Ironman Wisconsin finisher (Sep 2026): trained ~20 hours/week for 8 months alongside a full CS course load